Robotic Systems-2026

LAB 1 REPORT

Integrators: Forward Euler, Backward Euler, Runge-Kutta method

Team members:

Vladimir Toporkov, Anas Khmais Hamrouni

1. Introduction

In this lab, we are going to implement different types of integrators on a simple mechanical system. Together, we will cover all the methods for a linear case and your task is to perform the same procedure but for a nonlinear case.

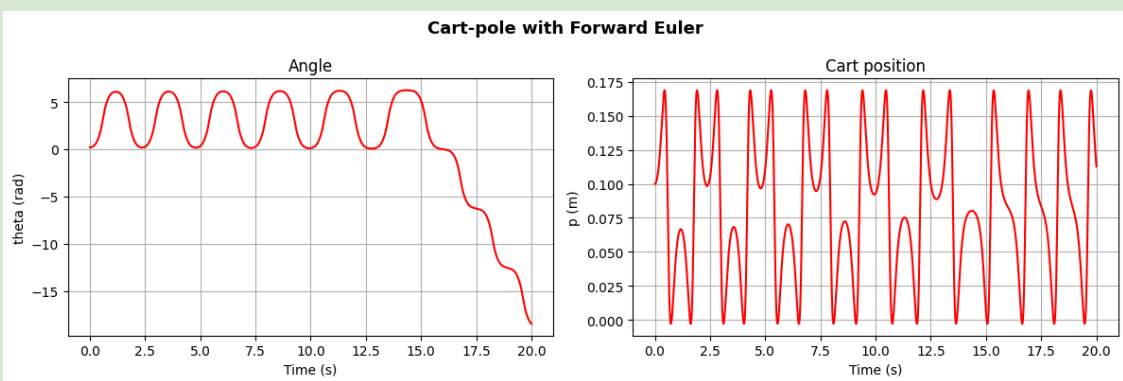
Forward Euler
Backward Euler
Runge-Kutta method

- You are free to choose the language to use (Python, C++, Matlab etc.)
- Also you can work in teams

Before starting the work, you can revise each of the methods in colab.

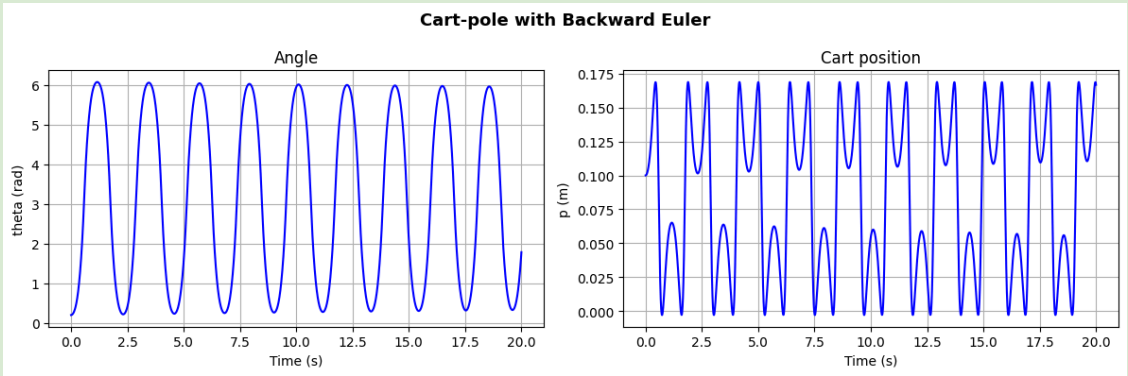
2. Forward Euler, Backward Euler, Runge-Kutta Method for **nonlinear** systems

Tasks:
2.1. Implement the Forward Euler technique for the cart pole system and include the response below.



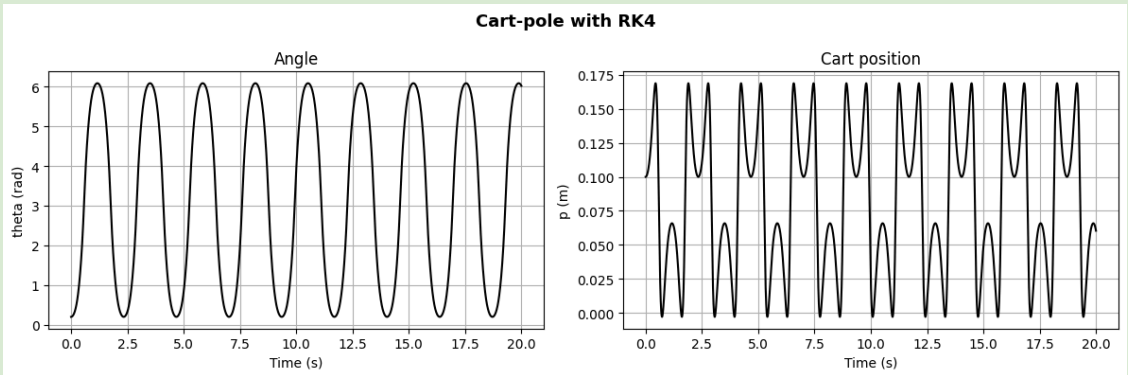
Plot for Task 2.1

2.2. Implement the Backward Euler technique for the given system and include the response below.



Plot for Task 2.2

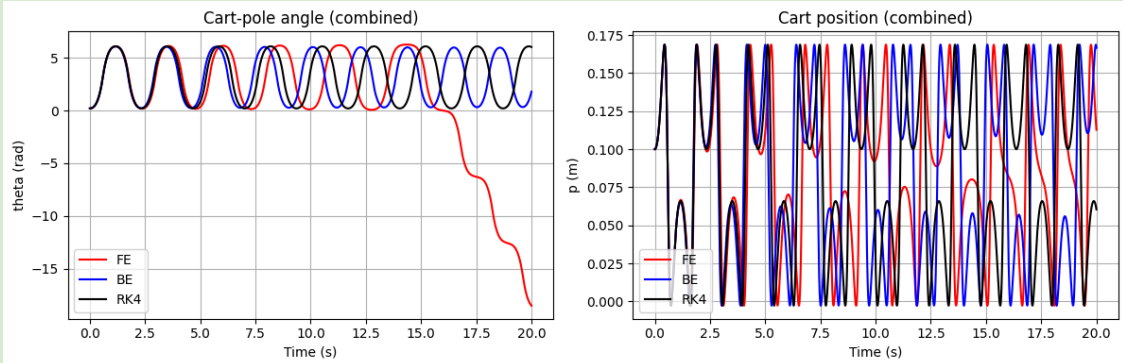
2.3. Implement [Runge-Kutta]() method for the given system and include the response below.



Plot for Task 2.3

2.4. For comparison, provide the plot, containing responses for all of the three methods above. Play with the time step dt (you may apply the following values dt=0.4, 0.004, 0.001) and try to answer the following questions:

- Which of the methods makes the system converge?
- Which of the methods makes the system diverge?

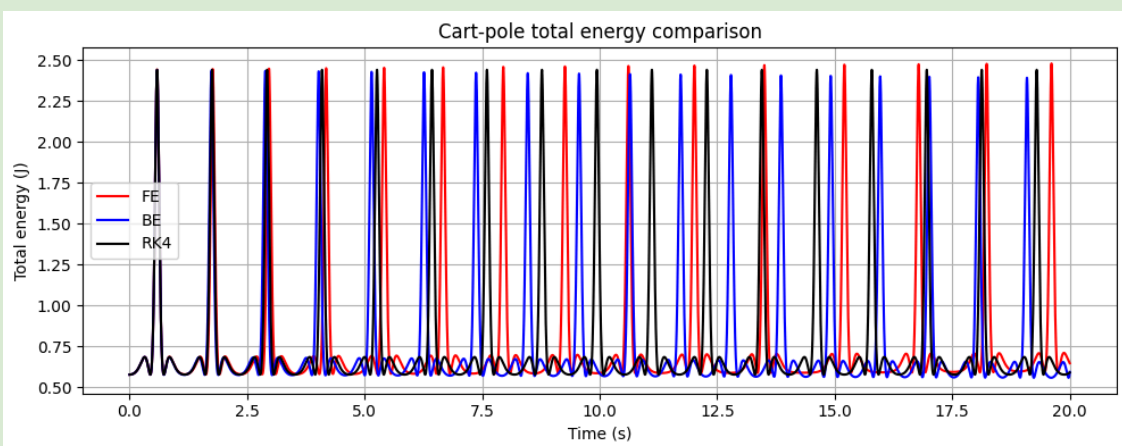


Plot for Task 2.4

Answer(s):
Forward Euler diverges, while Backward Euler and RK4 remain bounded (no clear convergence).

2.5. For each method count the time consumed on calculation. Which is the fastest?
Answer(s):
Forward Euler : 0.4264 s
Backward Euler : 2.2605 s
RK4:1.7702s
Forward Euler was the fastest

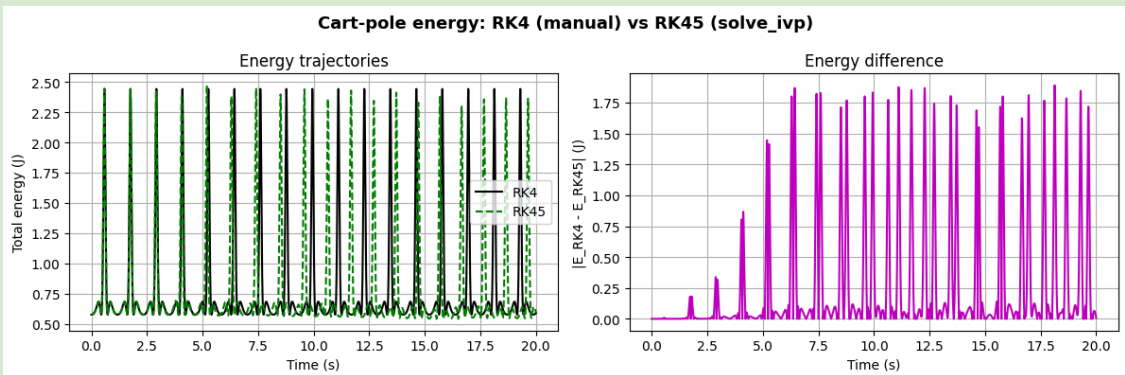
2.6. For each of the methods, plot the total energy of the system and compare them. How does the energy behave?



Plot for Task 2.6

Answer(s):
All three show oscillatory energy traces, but FE is clearly the least energy-consistent, while BE and especially RK4 preserve energy much better over the full simulation window.

2.7. Compare the Runge-Kutta method implemented and ready solver solver_ivp. Plot the energy for both of the methods. Are there any differences between them?



Plot for Task 2.7

Answer(s):
There are clear differences. For this cart-pole setup, manual RK4 preserves energy slightly better (smaller net drift), while RK45 (solve_ivp) diverges more from RK4 over long simulation time in both state and instantaneous energy.

2.8. (**At Home**) Function fsolve provides the solution for (nonlinear) equations given initial guess. Your task is to understand how this algorithm works by implementing the code from scratch (see Newton's method). As an example, you may take some well-known function, for example, $ax^2 + bx + c = 0$ and solve it by using the algorithm written by you.

Newton method results:
start 4.00 -> root 3.000000000000 in 6 iterations
start -2.00 -> root -0.500000000000 in 6 iterations
same roots as fsolve.